Experiment-3

Design a simple network topology using cisco packet tracer

Aim:

To design and configure the simple network topology with one router, two switches and multiple pcs using cisco packet tracer

Requirements:

cisco packet tracer 8x1, Router (2811 or 1841)

2 switches (2960), 4-6 PCs Ethernet cables

Procedure:

1) open cisco packet tracer and create new project

2) Drag 1 router, 2 switches & 4pc onto the work shop

3) connect pc0 & pc1 with switch0 using copper straight-through cable

4) connect pc2 & pc3 with switch1 using copper straight-through cable

5) connect switch0 to router Gb Ethernet 0/0

6) connect switch1 to router Gb ethernet 0/1

commands

Router > enable

Router # config terminal

Router (config) # hostname main router

main router (config) # interface gb ethernet 0/1

main router (config) # ip address 192.168.1.1 255.255.25

main router (config) # no shut down

main router (config-if) # exit

main router (config-if) # interface gb ethernet 0/1

main router (config-if) # ip address 192.168.2.1 255.255.255.0

main router (config-if) # no shut down

main router (config-if) # end

main router ~~ctrl~~ show ip interface brief

main router # show ip route

main router # copy running-config start up-config

7) config pc0 = ip= 192.168.1.2 , mask= 255.255.255.0

Gw = 192.168.1.1

8) config pc1 : ip= 192.168.1.3 mask = 255.255.255.0

Gw = 192.168.1.1.

9) config pc2 : ip= 192.168.2.2 mask = 255.255.255.0

Gw = 192.168.2.1

10) config pc3 : ip= 192.168.2.3 mask = 255.255.255.0

Gw = 192.168.2.2

11) config router int gol : ip address 192.168.2.1

255.255.255.0 -> no shutdown

Result:

A network topology with 1 router, 2 switch, and 4 PC's was successfully created inter network communication between the 2 subnets was verified using ping commands through cisco packet tracer.